

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science and Physics

Total Marks: 65

Time Allowed: 2 hours 30 minutes

Version: 1

General Instructions:

1. This paper has **three sections**: A (MCQs), B (Short Questions), C (Long Questions).
2. **Section A** (12 marks): 12 MCQs. Mark the correct bubble. Time: 15 minutes.
3. **Section B** (33 marks): 11 short questions $\times$ 3 marks each. Answer *either* the main question *or* the OR alternative.
4. **Section C** (20 marks): 4 long questions $\times$ 5 marks each. Answer *either* the main question *or* the OR alternative. Use $g = 10 \text{ m/s}^2$ in numerical problems unless otherwise specified.

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks) [15 min]

Q.1: Choose the correct answer and fill in the corresponding bubble.

#	Question	A	B	C	D	Answer
1	Which instrument is best suited to measure the length of a 2 m curtain rod?	Vernier caliper	Screw gauge	Measuring tape	Physical balance	☐ A ☐ B ☐ C ☐ D
2	A scale that always reads 2 g too high due to a calibration fault is affected by a:	Random error	Systematic error	Parallax error	Rounding error	☐ A ☐ B ☐ C ☐ D
3	The regular back-and-forth swinging of a pendulum bob is an example of:	Rotatory motion	Vibratory motion	Random motion	Translatory motion	☐ A ☐ B ☐ C ☐ D
4	A bus decelerates uniformly from 18 m/s to rest in 4 s. Its deceleration is:	4.5 m/s^2	0.22 m/s^2	22 m/s^2	72 m/s^2	☐ A ☐ B ☐ C ☐ D
5	A rocket accelerates upward by expelling gas downward. This is explained by:	Newton's first law	Newton's second law	Newton's third law	Conservation of mass	☐ A ☐ B ☐ C ☐ D
6	A trolley of mass 15 kg moves with a velocity of 4 m/s. Its momentum is:	$60 \text{ kg}\cdot\text{m/s}$	$3.75 \text{ kg}\cdot\text{m/s}$	$19 \text{ kg}\cdot\text{m/s}$	$30 \text{ kg}\cdot\text{m/s}$	☐ A ☐ B ☐ C ☐ D
7	Two people push a stalled car in the same direction. Their forces are:	Unlike parallel forces	Like parallel forces	Perpendicular forces	Balanced forces	☐ A ☐ B ☐ C ☐ D
8	A force of 45 N acts at a perpendicular distance of 0.6 m from a pivot. The torque produced is:	$27 \text{ N}\cdot\text{m}$	$75 \text{ N}\cdot\text{m}$	$0.013 \text{ N}\cdot\text{m}$	$45.6 \text{ N}\cdot\text{m}$	☐ A ☐ B ☐ C ☐ D
9	A spring extends by 10 cm when a load of 40 N is applied to it. Its spring constant is:	400 N/m	4 N/m	40 N/m	4000 N/m	☐ A ☐ B ☐ C ☐ D
10	A force of 150 N acts on an area of 0.03 m^2 . The pressure produced is:	5000 Pa	500 Pa	4.5 Pa	$50,000 \text{ Pa}$	☐ A ☐ B ☐ C ☐ D
11	Which renewable resource harnesses sea-level rise and fall caused by the Moon's gravity?	Geothermal	Tidal	Biomass	Wave	☐ A ☐ B ☐ C ☐ D
12	A statement that can never be tested or proven wrong by any experiment is:	A scientific law	A theory	Falsifiable	Unfalsifiable	☐ A ☐ B ☐ C ☐ D

SECTION B — Short Questions (11 × 3 = 33 Marks)

Q.2: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Part	Question	Marks	OR	OR Question	Marks
(i)	What instrument would you use to measure a 12 m school corridor, and why is it better suited than a metre rule? State one precaution against parallax error .	3	OR	Define least count . What is the least count of a metre rule graduated in millimetres? Is a reading of 1.234 m from it realistic? State how many decimal places she should report.	3
(ii)	Compare a vernier caliper and a screw gauge : which has the smaller least count, and what type of object is each best suited to measure?	3	OR	A screw gauge has pitch 0.5 mm and 25 divisions. Calculate its least count . If the main scale reads 2 mm and the 15th circular division coincides with the datum line, find the wire's diameter .	3
(iii)	A motorbike at 9 m/s accelerates uniformly at 2 m/s ² for 5 s. Calculate (a) final velocity, (b) distance covered, using $s = ut + \frac{1}{2}at^2$.	3	OR	A ball is dropped from a 180 m tower. Taking $g = 10$ m/s ² , calculate (a) time to reach the ground, (b) velocity just before impact.	3
(iv)	A cyclist accelerates from rest to 8 m/s in 4 s, then moves at this speed for 10 s more. What does the area under each part of the speed-time graph represent? Find the total distance covered.	3	OR	Define distance and displacement. A hiker walks 5 km north, 12 km east, then 5 km south. Calculate (a) total distance, (b) net displacement.	3
(v)	State Newton's second law ($F = ma$). A net force of 36 N acts on a 6 kg crate at rest. Calculate (a) its acceleration, (b) its velocity after 4 s.	3	OR	Define inertia . Why does a standing bus passenger fall backward when the bus suddenly starts moving forward?	3
(vi)	Define momentum and impulse. An airbag brings a 70 kg crash-test dummy from 15 m/s to rest in 0.2 s. Calculate (a) change in momentum, (b) average force exerted by the airbag.	3	OR	State the law of conservation of momentum. Skaters A (60 kg) and B (45 kg) push apart from rest on frictionless ice. If A moves at 3 m/s, find B's velocity and verify total momentum is zero.	3
(vii)	Define Optics , Classical Mechanics and Quantum Mechanics , giving one real-world example of each.	3	OR	Differentiate science , technology and engineering , using smartphone development as an example of each field's role.	3
(viii)	Differentiate like parallel forces and unlike parallel forces , with one everyday example of each.	3	OR	State the principle of moments . A metre rule pivoted at 35 cm has a 6 N weight at 10 cm. Where must a 5 N weight hang to balance it?	3
(ix)	State Pascal's principle . How does it apply to a hydraulic wheelchair lift raised by a small pedal force?	3	OR	Differentiate a hydraulic lift from a hydraulic car brake system: what each multiplies, and one similarity/difference in construction.	3
(x)	Define pressure and its SI unit. A wheelbarrow wheel exerts 900 N over a contact area of 0.006 m ² . Calculate the pressure exerted.	3	OR	Using $P = F/A$, explain why stiletto heels sink into soft ground while snow shoes do not, for similar body weight.	3
(xi)	Explain how biomass energy is obtained and used as fuel, naming two forms it can take.	3	OR	Explain how geothermal energy generates electricity, and state one advantage over fossil fuels.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Q.3: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	(a) Derive $s = ut + \frac{1}{2}at^2$ using a speed-time graph. [2] (b) A car at 12 m/s accelerates uniformly to 24 m/s in 8 s. Find (i) acceleration, (ii) distance covered. [2] (c) Verify (b)(ii) using $v^2 = u^2 + 2as$. [1]	5	OR	(a) State the three equations of motion, defining each symbol. [2] (b) A ball is thrown upward at 18 m/s. Taking $g = 10$ m/s ² , find the time to reach maximum height. [1] (c) Find the maximum height, and verify using $v^2 = u^2 - 2gh$. [2]	5
3(ii)	(a) State the law of conservation of momentum for an isolated system. [1] (b) A 4 kg rifle fires a 0.02 kg bullet at 400 m/s. Find the rifle's recoil velocity. [2] (c) If the bullet is in the barrel for 0.005 s, find the average force on it, and (by Newton's third law) on the rifle. [2]	5	OR	(a) State how momentum conservation applies to an explosion of a stationary body into two fragments. [1] (b) A stationary 10 kg bomb explodes into a 4 kg fragment at 15 m/s and a 6 kg fragment. Find the second fragment's velocity. [2] (c) Find the total kinetic energy released. [2]	5
3(iii)	(a) State Pascal's principle and explain its use in a hydraulic press that compacts scrap metal. [2] (b) A scrap-metal press has pistons of area 0.0025 m ² and 0.6 m ² . For an 80 N input force, find the compacting force delivered. [3]	5	OR	(a) Define a manometer and the principle behind its use. [2] (b) A U-tube manometer (liquid density 950 kg/m ³) shows a 32 cm height difference across a gas supply. Taking atmospheric pressure as 1.0×10^5 Pa and $g = 10$ m/s ² , find (i) the pressure difference, (ii) the absolute gas pressure. [3]	5
3(iv)	(a) State the law of conservation of energy. [1] (b) A 6 kg object is dropped from rest from 20 m. Taking $g = 10$ m/s ² , find (i) PE at the top, (ii) KE and speed just before impact. [4]	5	OR	(a) Differentiate renewable and non-renewable resources, naming two renewables besides solar, wind and hydro. [2] (b) A tidal barrage drops 5000 kg of water through 8 m every second. Taking $g = 10$ m/s ² , find (i) PE released/s, (ii) power generated. [3]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science and Physics

— End of Question Paper —